EE114 Homework 3

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Responses

1. **Question 2.5.2**

Question: Voice calls cost 20 cents each and data calls cost 30 cents each. C is the cost of one telephone call. The probability that a call is a voice call is $P[V] = 0.6$. The probability of a data call is $P[D] = 0.4$.

(a) Find $P_C(c)$, the PMF of C.

(b) What is $E[C]$, the expected value of C?

Response:

(a) Find $P_C(c)$, the PMF of C. The random variable C maps the outcome of the call type to its cost. If the call is a voice call, $C = 20$. If the call is a data call, $C = 30$. The Probability Mass Function (PMF), $P_C(c)$, is the probability that the random variable C takes on a specific value c.

$$P_C(20) = P[V] = 0.6, \qquad P_C(30) = P[D] = 0.4,$$

and $P_C(c) = 0$ for all other values.

Answer: The PMF can be written as

$$P_C(c) = \begin{cases} 0.6, & \text{for } c = 20, \\ 0.4, & \text{for } c = 30, \\ 0, & \text{otherwise.} \end{cases}$$

(b) What is $E[C]$, the expected value of C? The expected value $E[C]$ is the probability-weighted average of all possible values.

$$E[C] = \sum_{c} c \cdot P_C(c).$$

Substituting the values from the PMF:

$$E[C] = (20 \cdot 0.6) + (30 \cdot 0.4) = 12 + 12 = 24.$$

Answer: The expected value of C is 24 cents.

2. **Question 2.5.4**

Question: Find the expected value of the random variable X in Problem 2.4.2. (For reference, the random variable X from Problem 2.4.2 is defined by the following CDF:)

$$F_X(x) = \begin{cases} 0, & x < -1, \\ 0.2, & -1 \leq x < 0, \\ 0.7, & 0 \leq x < 1, \\ 1, & x \geq 1. \end{cases}$$

Response:

At $x = -1$: The CDF jumps from 0 to 0.2.

$$P_X(-1) = 0.2 - 0 = 0.2.$$

At $x = 0$: The CDF jumps from 0.2 to 0.7.

$$P_X(0) = 0.7 - 0.2 = 0.5.$$

At $x = 1$: The CDF jumps from 0.7 to 1.

$$P_X(1) = 1 - 0.7 = 0.3.$$

So, the PMF is:

$$P_X(-1) = 0.2, \qquad P_X(0) = 0.5, \qquad P_X(1) = 0.3.$$

Step 2: Calculate the expected value $E[X]$. The expected value is the sum of each value multiplied by its probability:

$$E[X] = \sum_x x \cdot P_X(x).$$

Substituting our values:

$$E[X] = (-1 \cdot 0.2) + (0 \cdot 0.5) + (1 \cdot 0.3).$$

Calculate the terms:

$$E[X] = -0.2 + 0 + 0.3 = 0.1.$$

Answer: The expected value of the random variable X is 0.1.

3. **Question 2.5.9**

Question: Suppose you go to a casino with exactly \$63. At this casino, the only game is roulette and the only bets allowed are red and green. In addition, the wheel is fair so that $P[\text{red}] = P[\text{green}] = 1/2$. You have the following strategy: First, you bet \$1. If you win the bet, you leave the casino with \$64. If you lose, you then bet \$2. If you win, you quit and go home. If you lose, you bet \$4. In fact, whenever you lose, you double your bet until either you win a bet or you lose all of your money. However, as soon as you win a bet, you quit and go home. Let Y equal the amount of money that you take home. Find $P_Y(y)$ and $E[Y]$. Would you like to play this game every day?

Response:

1. Find $P_Y(y)$, the PMF of Y

You begin with \$63. The strategy of doubling bets (Martingale) allows for a maximum sequence of 6 bets $(1 + 2 + 4 + 8 + 16 + 32 = 63)$.

Case 1: You lose everything ($Y = 0$).

This occurs only if you lose 6 consecutive bets.

$$P(Y = 0) = \left(\frac{1}{2}\right)^6 = \frac{1}{64}$$

Case 2: You win ($Y = 64$).

This occurs if you win any single bet before running out of money.

$$P(Y = 64) = 1 - P(Y = 0) = 1 - \frac{1}{64} = \frac{63}{64}$$

The PMF is:

$$P_Y(y) = \begin{cases} \frac{63}{64} & y = 64 \\ \frac{1}{64} & y = 0 \\ 0 & \text{otherwise} \end{cases}$$

2. Find $E[Y]$

$$E[Y] = \sum y \cdot P_Y(y)$$

$$E[Y] = \left(64 \cdot \frac{63}{64}\right) + \left(0 \cdot \frac{1}{64}\right)$$

$$E[Y] = 63 + 0$$

$$E[Y] = 63$$

3. Would you like to play this game every day?

No. Even though the probability of winning is high (98.4%), the expected profit is zero ($E[Y] = \$63$, which is exactly what you started with). You are risking your entire bankroll of $63 to gain just $1. Over time, the single catastrophic loss of $63 (which happens on average once every 64 days) will wipe out all the small $1 wins accumulated.

4. **Question 3.1.1**

Question: The cumulative distribution function of random variable X is

$$F_X(x) = \begin{cases} 0, & x < -1, \\ (x+1)/2, & -1 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

(a) What is $P[X > 1/2]$?

(b) What is $P[-1/2 < X \le 3/4]$?

(c) What is $P[|X| \le 1/2]$?

(d) What is the value of a such that $P[X \le a] = 0.8$?

Response:

(a) **What is $P[X > 1/2]$?**

Using the property $P[X > x] = 1 - P[X \le x] = 1 - F_X(x)$:

$$P[X > 1/2] = 1 - F_X(1/2)$$

Since $-1 \le 1/2 < 1$, we use the formula $\frac{x+1}{2}$:

$$F_X(1/2) = \frac{0.5 + 1}{2} = \frac{1.5}{2} = 0.75$$

$$P[X > 1/2] = 1 - 0.75 = 0.25$$

Answer: 0.25

(b) **What is $P[-1/2 < X \le 3/4]$?**

Using the property $P[a < X \le b] = F_X(b) - F_X(a)$:

$$P[-1/2 < X \le 3/4] = F_X(3/4) - F_X(-1/2)$$

Calculate $F_X(3/4)$:

$$F_X(3/4) = \frac{0.75 + 1}{2} = \frac{1.75}{2} = 0.875$$

Calculate $F_X(-1/2)$:

$$F_X(-1/2) = \frac{-0.5+1}{2} = \frac{0.5}{2} = 0.25$$

Subtract the two:

$$P[-1/2 < X \leq 3/4] = 0.875 - 0.25 = 0.625$$

Answer: 0.625

(c) **What is $P[|X| \leq 1/2]$?**

The inequality $|X| \leq 1/2$ is equivalent to $-1/2 \leq X \leq 1/2$.

$$P[-1/2 \leq X \leq 1/2] = F_X(1/2) - F_X(-1/2)$$

Calculate $F_X(1/2)$:

$$F_X(1/2) = \frac{0.5+1}{2} = 0.75$$

Calculate $F_X(-1/2)$:

$$F_X(-1/2) = \frac{-0.5+1}{2} = 0.25$$

Subtract the two:

$$P[|X| \leq 1/2] = 0.75 - 0.25 = 0.5$$

Answer: 0.5

(d) **What is the value of a such that $P[X \leq a] = 0.8$?**

We are looking for a where $F_X(a) = 0.8$. Since $0 \leq 0.8 \leq 1$, the value a must fall within the range $[-1, 1)$. We use the middle part of the function definition:

$$\frac{a+1}{2} = 0.8$$

Multiply both sides by 2:

$$a + 1 = 1.6$$

Subtract 1 from both sides:

$$a = 0.6$$

Answer: $a = 0.6$

5. **Question 3.1.2**

Question: The cumulative distribution function of the continuous random variable V is

$$F_V(v) = \begin{cases} 0, & v < -5, \\ c(v+5)^2, & -5 \leq v < 7, \\ 1, & v \geq 7. \end{cases}$$

(a) What is c?

(b) What is $P[V > 4]$?

(c) $P[-3 < V \leq 0]$?

(d) What is the value of a such that $P[V > a] = 2/3$?

Response:

(a) **What is c?**

For a valid CDF, the function must be continuous (no jumps) and approach 1 at the maximum value of the range. Therefore, at the upper boundary $v = 7$, $F_V(7)$ must equal 1.

$$c(7+5)^2 = 1$$

$$c(12)^2 = 1$$

$$144c = 1$$

$$c = \frac{1}{144}$$

(b) **What is $P[V > 4]$?**

Using the complement rule: $P[V > 4] = 1 - P[V \leq 4] = 1 - F_V(4)$. First, calculate $F_V(4)$ using $c = \frac{1}{144}$:

$$F_V(4) = \frac{1}{144}(4+5)^2$$

$$F_V(4) = \frac{1}{144}(9)^2$$

$$F_V(4) = \frac{81}{144}$$

Simplify the fraction (divide by 9):

$$F_V(4) = \frac{9}{16}$$

Now, find the probability:

$$P[V > 4] = 1 - \frac{9}{16} = \frac{7}{16}$$

Answer: $\frac{7}{16}$

(c) **What is** $P[-3 < V \leq 0]$**?**

The probability for an interval is $F_V(b) - F_V(a)$.

$$P[-3 < V \leq 0] = F_V(0) - F_V(-3)$$

Calculate $F_V(0)$:

$$F_V(0) = \frac{1}{144}(0+5)^2 = \frac{25}{144}$$

Calculate $F_V(-3)$:

$$F_V(-3) = \frac{1}{144}(-3+5)^2 = \frac{1}{144}(2)^2 = \frac{4}{144}$$

Subtract the two:

$$P[-3 < V \leq 0] = \frac{25}{144} - \frac{4}{144} = \frac{21}{144}$$

Simplify the fraction (divide by 3):

$$\frac{21}{144} = \frac{7}{48}$$

Answer: $\frac{7}{48}$

(d) **What is the value of** a **such that** $P[V > a] = 2/3$**?**

We know that $P[V > a] = 1 - F_V(a)$. So:

$$1 - F_V(a) = \frac{2}{3}$$

$$F_V(a) = \frac{1}{3}$$

Substitute the formula for $F_V(a)$:

$$\frac{1}{144}(a+5)^2 = \frac{1}{3}$$

Multiply both sides by 144:

$$(a+5)^2 = \frac{144}{3}$$

$$(a+5)^2 = 48$$

Take the square root of both sides (we take the positive root because a must be greater than -5):

$$a+5 = \sqrt{48}$$

$$a+5 = \sqrt{16 \cdot 3}$$

$$a+5 = 4\sqrt{3}$$

Solve for a:

$$a = 4\sqrt{3} - 5$$

(Note: $\sqrt{3} \approx 1.732$, so $a \approx 1.93$, which is within the valid range of -5 to 7.)
Answer: $a = 4\sqrt{3} - 5$

6. **Question 3.2.1**

Question: The random variable X has probability density function (PDF)

$$f_X(x) = \begin{cases} cx, & 0 \le x \le 2, \\ 0, & \text{otherwise.} \end{cases}$$

Use the PDF to find

(a) the constant c,

(b) $P[0 \le X \le 1]$,

(c) $P[-1/2 \le X \le 1/2]$,

(d) the CDF $F_X(x)$. $\rightarrow \frac{1}{2}x$

Response:

(a) **Find the constant c.**

A fundamental property of any PDF is that the total area under the curve must equal 1.

$$\int_{-\infty}^{\infty} f_X(x)\,dx = 1$$

Substitute the given function:

$$\int_0^2 cx\,dx = 1$$

Evaluate the integral:

$$\left[\frac{cx^2}{2}\right]_0^2 = 1$$

$$\frac{c(2)^2}{2} - \frac{c(0)^2}{2} = 1$$

$$\frac{4c}{2} = 1$$

$$2c = 1$$

$$c = \frac{1}{2}$$

(b) **Find $P[0 \le X \le 1]$.**

To find the probability over an interval, integrate the PDF over that interval. Using

$c = 1/2$:

$$P[0 \leq X \leq 1] = \int_0^1 \frac{1}{2} x \, dx$$
$$= \left[\frac{x^2}{4}\right]_0^1$$
$$= \frac{1^2}{4} - \frac{0^2}{4}$$
$$= \frac{1}{4}$$

Answer: 0.25 (or 1/4)

(c) **Find** $P[-1/2 \leq X \leq 1/2]$**.**

The PDF is non-zero only for $x \geq 0$. Therefore, the probability for the range $[-1/2, 0)$ is 0. We effectively calculate the probability from 0 to 1/2.

$$P[-1/2 \leq X \leq 1/2] = \int_{-1/2}^{1/2} f_X(x) \, dx$$
$$= \int_{-1/2}^{0} 0 \, dx + \int_0^{1/2} \frac{1}{2} x \, dx$$
$$= 0 + \left[\frac{x^2}{4}\right]_0^{1/2}$$
$$= \frac{(1/2)^2}{4} - 0$$
$$= \frac{1/4}{4}$$
$$= \frac{1}{16}$$

Answer: 0.0625 (or 1/16)

(d) **Find the CDF** $F_X(x)$**.**

The Cumulative Distribution Function (CDF) is defined as $F_X(x) = \int_{-\infty}^{x} f_X(t) \, dt$. We calculate this for each region:

For $x < 0$**:** The PDF is 0, so the area is 0.

$$F_X(x) = 0$$

For $0 \leq x \leq 2$**:** We integrate from the start of the support (0) to x.

$$F_X(x) = \int_0^x \frac{1}{2}t\,dt$$

$$F_X(x) = \left[\frac{t^2}{4}\right]_0^x = \frac{x^2}{4}$$

For $x > 2$**:** The total probability has been accumulated.

$$F_X(x) = 1$$

Answer:

$$F_X(x) = \begin{cases} 0 & x < 0 \\ \frac{x^2}{4} & 0 \leq x \leq 2 \\ 1 & x > 2 \end{cases}$$

7. **Question 3.2.2**

Question: The cumulative distribution function of random variable X is

$$F_X(x) = \begin{cases} 0, & x < -1, \\ (x+1)/2, & -1 \le x < 1, \\ 1, & x \ge 1. \end{cases}$$

Find the PDF $f_X(x)$ of X.

Response:

We differentiate $F_X(x)$ piece by piece for each interval:

For $x < -1$:

$$F_X(x) = 0 \implies \frac{d}{dx}(0) = 0$$

For $-1 < x < 1$:

$$F_X(x) = \frac{x+1}{2} = \frac{1}{2}x + \frac{1}{2}$$

$$\frac{d}{dx}\left(\frac{1}{2}x + \frac{1}{2}\right) = \frac{1}{2}$$

For $x > 1$:

$$F_X(x) = 1 \implies \frac{d}{dx}(1) = 0$$

(Note: Since the CDF is continuous at the boundaries $x = -1$ and $x = 1$, there are no impulses (delta functions) in the PDF.)

Answer: Combining these results, the PDF is:

$$f_X(x) = \begin{cases} \frac{1}{2}, & -1 \le x < 1, \\ 0, & \text{otherwise.} \end{cases}$$

8. **Question 3.2.3**

Question: Find the PDF $f_U(u)$ of the random variable U in Problem 3.1.3.

(Note: In the original text, Problem 3.1.3 likely defines a random variable labeled W. Since 3.2.3 asks for the random variable U, it is generally assumed that U is the same variable as W from the previous problem. I will solve for the variable W.)

Reference Problem 3.1.3 (Given Information): The Cumulative Distribution Function (CDF) of the random variable W is:

$$F_W(w) = \begin{cases} 0, & w < -5, \\ \frac{w+5}{8}, & -5 \leq w < -3, \\ \frac{1}{4}, & -3 \leq w < 3, \\ \frac{1}{4} + \frac{3(w-3)}{8}, & 3 \leq w < 5, \\ 1, & w \geq 5. \end{cases}$$

Response:

We differentiate $F_W(w)$ in each interval:

For $w < -5$:

$$F_W(w) = 0 \implies f_W(w) = 0$$

For $-5 \leq w < -3$:

$$F_W(w) = \frac{w+5}{8} = \frac{1}{8}w + \frac{5}{8}$$

$$f_W(w) = \frac{d}{dw}\left(\frac{1}{8}w + \frac{5}{8}\right) = \frac{1}{8}$$

For $-3 \leq w < 3$:

$$F_W(w) = \frac{1}{4}$$

$$f_W(w) = \frac{d}{dw}\left(\frac{1}{4}\right) = 0$$

For $3 \leq w < 5$:

$$F_W(w) = \frac{1}{4} + \frac{3(w-3)}{8} = \frac{1}{4} + \frac{3}{8}w - \frac{9}{8}$$

$$f_W(w) = \frac{d}{dw}\left(\frac{3}{8}w + \text{constant}\right) = \frac{3}{8}$$

For $w \geq 5$**:**

$$F_W(w) = 1 \implies f_W(w) = 0$$

Answer: The PDF is:

$$f_W(w) = \begin{cases} \frac{1}{8}, & -5 \leq w < -3, \\ 0, & -3 \leq w < 3, \\ \frac{3}{8}, & 3 \leq w < 5, \\ 0, & \text{otherwise.} \end{cases}$$

9. **Question 3.3.1 Question:** Continuous random variable X has PDF

$$f_X(x) = \begin{cases} 1/4, & -1 \leq x \leq 3, \\ 0, & \text{otherwise.} \end{cases}$$

Define the random variable Y by $Y = h(X) = X^2$.

(a) Find $E[X]$ and $\text{Var}[X]$.

(b) Find $h(E[X])$ and $E[h(X)]$.

(c) Find $E[Y]$ and $\text{Var}[Y]$.

Response:

(a) **Find $E[X]$ and Var$[X]$.**

Since X is a uniform random variable on the interval $[a, b]$ where $a = -1$ and $b = 3$, we can use the standard formulas for the mean and variance of a uniform distribution.

Expected Value $E[X]$:

$$E[X] = \frac{a+b}{2} = \frac{-1+3}{2} = \frac{2}{2} = 1$$

Variance Var$[X]$:

$$\text{Var}[X] = \frac{(b-a)^2}{12} = \frac{(3-(-1))^2}{12} = \frac{4^2}{12} = \frac{16}{12} = \frac{4}{3}$$

Answer: $E[X] = 1$ $\text{Var}[X] = 4/3$

(b) **Find $h(E[X])$ and $E[h(X)]$.**

1. **Calculate $h(E[X])$:** This is the function h evaluated at the expected value of X. From part (a), $E[X] = 1$. Since $h(x) = x^2$:

$$h(E[X]) = (E[X])^2 = (1)^2 = 1$$

2. **Calculate $E[h(X)]$:** This is the expected value of the random variable X^2. We can calculate this using the formula $\text{Var}[X] = E[X^2] - (E[X])^2$, which rearranges

to:

$$E[X^2] = \text{Var}[X] + (E[X])^2$$

Substituting the values from part (a):

$$E[X^2] = \frac{4}{3} + (1)^2 = \frac{4}{3} + 1 = \frac{7}{3}$$

(Alternatively, you can compute it via integration: $\int_{-1}^{3} x^2 \cdot \frac{1}{4} dx = \frac{7}{3}$)

Answer: $h(E[X]) = 1$ $E[h(X)] = 7/3$

(c) **Find $E[Y]$ and Var$[Y]$.**

Since $Y = X^2$, the term $E[Y]$ is exactly what we calculated in part (b) as $E[h(X)]$.

1. **Calculate $E[Y]$:**

$$E[Y] = E[X^2] = \frac{7}{3}$$

2. **Calculate Var$[Y]$:** To find the variance of Y, we use the standard variance formula:

$$\text{Var}[Y] = E[Y^2] - (E[Y])^2$$

First, we need to find $E[Y^2]$. Since $Y = X^2$, then $Y^2 = (X^2)^2 = X^4$.

$$E[Y^2] = E[X^4] = \int_{-\infty}^{\infty} x^4 f_X(x)\, dx$$

Substitute the PDF (1/4 between -1 and 3):

$$E[X^4] = \int_{-1}^{3} x^4 \cdot \frac{1}{4}\, dx$$

$$= \frac{1}{4}\left[\frac{x^5}{5}\right]_{-1}^{3}$$

$$= \frac{1}{20}\left(3^5 - (-1)^5\right)$$

$$= \frac{1}{20}(243 - (-1))$$

$$= \frac{244}{20} = \frac{61}{5}$$

Now, substitute $E[Y^2]$ and $E[Y]$ back into the variance formula:

$$\text{Var}[Y] = \frac{61}{5} - \left(\frac{7}{3}\right)^2$$

$$\text{Var}[Y] = \frac{61}{5} - \frac{49}{9}$$

Find a common denominator (45):

$$= \frac{61 \cdot 9}{45} - \frac{49 \cdot 5}{45}$$

$$= \frac{549}{45} - \frac{245}{45}$$

$$= \frac{304}{45}$$

Answer: $E[Y] = 7/3$ $\text{Var}[Y] = 304/45$ (approx 6.756)

10. **Question 3.3.2**

Question: Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} 1/8 & 1 \leq x \leq 9, \\ 0 & \text{otherwise.} \end{cases}$$

Let $Y = h(X) = 1/\sqrt{X}$.

(a) Find $E[X]$ and $\text{Var}[X]$.

(b) Find $h(E[X])$ and $E[h(X)]$.

(c) Find $E[Y]$ and $\text{Var}[Y]$.

Response:

(a) **Find $E[X]$ and Var$[X]$.**

Since X is a uniform random variable on the interval $[a, b]$ where $a = 1$ and $b = 9$:

Expected Value $E[X]$:

$$E[X] = \frac{a+b}{2} = \frac{1+9}{2} = \frac{10}{2} = 5$$

Variance Var$[X]$:

$$\text{Var}[X] = \frac{(b-a)^2}{12} = \frac{(9-1)^2}{12} = \frac{8^2}{12} = \frac{64}{12}$$

Simplify the fraction (divide by 4):

$$\text{Var}[X] = \frac{16}{3}$$

Answer: $E[X] = 5$ $\text{Var}[X] = 16/3$ (approx 5.33)

(b) **Find $h(E[X])$ and $E[h(X)]$.**

1. **Calculate $h(E[X])$:** This is the function h evaluated at the expected value of X (which is 5).

$$h(E[X]) = \frac{1}{\sqrt{E[X]}} = \frac{1}{\sqrt{5}}$$

2. **Calculate $E[h(X)]$:** This is the expected value of the transformed variable $1/\sqrt{X}$.

$$E[h(X)] = \int_{-\infty}^{\infty} h(x) f_X(x)\, dx$$

$$E[h(X)] = \int_{1}^{9} \frac{1}{\sqrt{x}} \cdot \frac{1}{8}\, dx$$

$$= \frac{1}{8} \int_{1}^{9} x^{-1/2}\, dx$$

Integrate using the power rule $\int x^n dx = \frac{x^{n+1}}{n+1}$:

$$= \frac{1}{8} \left[\frac{x^{1/2}}{1/2} \right]_1^9$$

$$= \frac{1}{8} \cdot 2 \left[\sqrt{x} \right]_1^9$$

$$= \frac{1}{4}(\sqrt{9} - \sqrt{1})$$

$$= \frac{1}{4}(3-1) = \frac{2}{4} = \frac{1}{2}$$

Answer: $h(E[X]) = 1/\sqrt{5}$ $E[h(X)] = 0.5$

(c) **Find $E[Y]$ and Var$[Y]$.**

Since $Y = h(X)$, $E[Y]$ is the same as $E[h(X)]$ calculated above.

1. **Calculate $E[Y]$:**

$$E[Y] = 0.5$$

2. **Calculate Var$[Y]$:** Use the formula $\text{Var}[Y] = E[Y^2] - (E[Y])^2$. First, find $E[Y^2]$. Note that $Y^2 = (1/\sqrt{X})^2 = 1/X$.

$$E[Y^2] = E[1/X] = \int_1^9 \frac{1}{x} \cdot \frac{1}{8}\, dx$$

$$= \frac{1}{8} \left[\ln(x) \right]_1^9$$

$$= \frac{1}{8}(\ln(9) - \ln(1))$$

Since $\ln(1) = 0$:

$$E[Y^2] = \frac{\ln(9)}{8}$$

Now substitute back into the variance formula:

$$\text{Var}[Y] = \frac{\ln(9)}{8} - (0.5)^2$$

$$\text{Var}[Y] = \frac{\ln(9)}{8} - 0.25$$

(Using approximations: $\ln(9) \approx 2.197$, so $E[Y^2] \approx 0.2746$. $\text{Var}[Y] \approx 0.2746 - 0.25 = 0.0246$)

Answer: $E[Y] = 0.5$ $\text{Var}[Y] = \frac{\ln(9)}{8} - \frac{1}{4}$

11. **Question 3.3.5**

Question: The cumulative distribution function of the random variable Y is

$$F_Y(y) = \begin{cases} 0, & y < -1, \\ (y+1)/2, & -1 \le y \le 1, \\ 1, & y > 1. \end{cases}$$

(a) What is $E[Y]$?

(b) What is $\text{Var}[Y]$?

Response:

Note that differentiating the CDF for $-1 \le y \le 1$ gives $f_Y(y) = \frac{d}{dy}(\frac{y+1}{2}) = \frac{1}{2}$, which corresponds to a uniform distribution on $[-1, 1]$.

(a) **What is $E[Y]$?**

Since Y is uniformly distributed on $[a, b]$ where $a = -1$ and $b = 1$, the expected value is simply the midpoint of the interval.

Method 1: Using the Uniform Distribution Formula

$$E[Y] = \frac{a+b}{2} = \frac{-1+1}{2} = \frac{0}{2} = 0$$

Method 2: Using Integration

$$E[Y] = \int_{-\infty}^{\infty} y \cdot f_Y(y)\, dy$$

$$E[Y] = \int_{-1}^{1} y \cdot \frac{1}{2}\, dy$$

$$E[Y] = \frac{1}{2}\left[\frac{y^2}{2}\right]_{-1}^{1}$$

$$E[Y] = \frac{1}{4}(1^2 - (-1)^2) = \frac{1}{4}(1-1) = 0$$

Answer: $E[Y] = 0$

(b) **What is $\text{Var}[Y]$?**

Method 1: Using the Uniform Distribution Formula For a uniform distribution on $[a, b]$, the variance is $\frac{(b-a)^2}{12}$.

$$\text{Var}[Y] = \frac{(1-(-1))^2}{12} = \frac{(2)^2}{12} = \frac{4}{12} = \frac{1}{3}$$

Method 2: Using the Definition $\text{Var}[Y] = E[Y^2] - (E[Y])^2$. Since $E[Y] = 0$, we just need to find $E[Y^2]$.

$$E[Y^2] = \int_{-1}^{1} y^2 \cdot \frac{1}{2}\, dy$$

$$E[Y^2] = \frac{1}{2}\left[\frac{y^3}{3}\right]_{-1}^{1}$$

$$E[Y^2] = \frac{1}{6}(1^3 - (-1)^3)$$

$$E[Y^2] = \frac{1}{6}(1-(-1)) = \frac{1}{6}(2) = \frac{1}{3}$$

So, $\text{Var}[Y] = \frac{1}{3} - 0^2 = \frac{1}{3}$.

Answer: $\text{Var}[Y] = 1/3$